

Element F

When group 6 first started brainstorming the idea of the Enchanted mirror, the members ranked each idea on consistency, cost, and design, and then voted on it. After voting, the group came up with the idea of making a mirror out of the wood provided by Ms. Moody and installing LEDs.

Ms. Moody's main requirements were that it was remote controlled and that it "looked good". The budget given was around 60 dollars. With the budget and requirements in mind, we decided to use the wood provided by the theater for the outside of the mirror and plexiglass provided by Mr. Kiser as the diffuser. After deciding this, we did research into cheap but well performing remote controlled LEDs. We looked into reviews and videos on how the LEDs performed and how they worked and decided to purchase some off amazon. After the LEDs arrived, we made sure they were remote controlled and that they had some decent range. Both were true and the range was about 30ft. After this we installed the LEDs into the mirror and tested it. We had to make a hole in the back of the mirror so that the LED sensor could be hit by the infrared light. After this, we brought it to the stage where we tested its longevity angles in which we would turn it on and distance again. We found the the battery source was not viable and ordered a battery pack with a on and off switch to ensure it would not go into power saving mode. After installing the the new pack, we determined that the mirror worked and got it checked by Ms. Moody who said she liked it.

Objective:

Make a magic mirror the looks good and can be turned on by a remote control

Procedure:

1. Open mirror
2. Install batteries in pack
3. Turn on the pack
4. Close mirror
5. Position yourself behind the mirror
6. Direct remote control at the back of the mirror
7. Hit the on button on the remote control
8. Adjust the colors
9. Direct remote control at the back of the mirror
10. When done hit the off button